S MUHILAN

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CAREER DESCRIPTION

Computer Science undergraduate specialising in Cloud Computing with hands-on experience in AWS, Python, Apache Spark, and AI-based analytics. Passionate about building scalable, intelligent systems that bridge cloud infrastructure and real-world applications. Demonstrated ability to deliver end-to-end projects spanning offline edge computing, machine learning-driven stock prediction, and large-scale data engineering. Seeking an internship or entry-level role where I can contribute to impactful cloud and AI solutions while continuing to grow as a software engineer.

EDUCATION

B.Tech – Computer Science & Engineering (Cloud Computing) *Expected 2027*
SRM Institute of Science and Technology – Ramapuram, Chennai | CGPA: 8.05 / 10
Class XII – State Board (PCMB) *2023*
Christ the King Boys Matric Hr Sec School, Kumbakonam | 71%

PROJECTS

Cloudster: Autonomous Edge-Education Node *Jan 2026*
Technologies: Golang, Node.js, Python, Raspberry Pi, Ollama, WebSockets

- Designed and deployed a self-contained edge computing platform on Raspberry Pi hardware, enabling AI-powered classroom services — including intelligent Q&A, content delivery, and interactive learning — in environments with zero internet connectivity.
- Engineered a real-time bidirectional communication layer using WebSockets to synchronise multiple client devices with the edge node, ensuring low-latency responses and seamless multi-user classroom experience.
- Integrated the Ollama framework to run large language models locally on the edge device, eliminating cloud dependency while maintaining conversational AI quality comparable to cloud-hosted solutions.
- Developed backend microservices in Golang for high-performance request handling and used Node.js for API orchestration, with Python powering the AI inference and data preprocessing pipelines.

Stock Vision AI with Apache Spark *Oct 2025*
Technologies: Apache Spark, Python, PySpark MLlib, Pandas, Matplotlib

- Architected a distributed data processing pipeline using Apache Spark to ingest, clean, and transform large-scale historical stock market datasets, significantly reducing preprocessing time compared to single-node approaches.
- Implemented machine learning models with PySpark MLlib, including regression-based trend forecasting and classification models for buy/sell signal generation, enabling data-driven investment insights.
- Built a real-time analytics module that streams live market data through Spark Structured Streaming, processing tick-by-tick price changes and generating rolling indicators such as moving averages and RSI.
- Created interactive visualisation dashboards using Matplotlib and Pandas to display model predictions, confidence intervals, and historical performance metrics for portfolio evaluation.

ML Stock Analysis Tool – Project Intern, Institution of Engineers (India) – Qatar Chapter *Oct 2025*
Technologies: Python, Yahoo Finance API, Linear Regression, Decision Tree, Matplotlib, Seaborn

- Developed a full-featured stock analysis application that fetches real-time and historical financial data via the Yahoo Finance API, applies machine learning models to identify trends, and generates actionable forecasts.
- Implemented and compared Linear Regression and Decision Tree algorithms for price prediction, performing feature engineering on OHLCV data including rolling averages, momentum indicators, and volume-weighted signals.
- Built an interactive visualisation dashboard using Matplotlib and Seaborn that presents price forecasts, model accuracy metrics, and comparative equity performance, enabling stakeholders to make informed decisions.

CERTIFICATIONS & ACHIEVEMENTS

- **– CDAC Chennai & SRM IST (Feb 2026)**Cloud Computing Bootcamp
- **– NPTEL (Jan 2025)**Introduction to Database Systems
- **– Participant, GDG VIT Chennai (Mar 2026)**DevsHouse '26 National Level Hackathon
- **– Attendee, NIT Tiruchirappalli (Jul 2025)**AI in Legal Domain Workshop
- **– District-level competitive player**Badminton

SKILLS & LANGUAGES

Programming	Python, SQL
Tools & Cloud	AWS, Docker, Git, GitHub, Apache Spark, AI-Assisted Development
Domain	Cloud Computing, Machine Learning, Edge Computing, Data Engineering
Languages	English (Professional Working Proficiency), Tamil (Native Proficiency)